

Student Registration Form – Dynamic Website on AWS EC2

About This Project

In this project, I created a **Student Registration Form** as a dynamic website where user details are stored in a MySQL database.

Earlier, I had only worked on static websites where data was not stored anywhere. So I wanted to understand how real applications work when a user submits a form.

In this project, when a user enters their details (name, email, password), the data is sent to a PHP file and then stored in the database. This helped me understand the complete flow from frontend to backend and database.

I also used CSS to design the form so that it looks clean and user-friendly instead of a plain basic form.

What I wanted to learn

- How form data is handled
- How PHP connects with MySQL
- How data is stored in database
- How to deploy a dynamic website on AWS EC2

✨ This is a beginner-level project, but it gave me a clear idea of how real websites manage user data.

What I Built

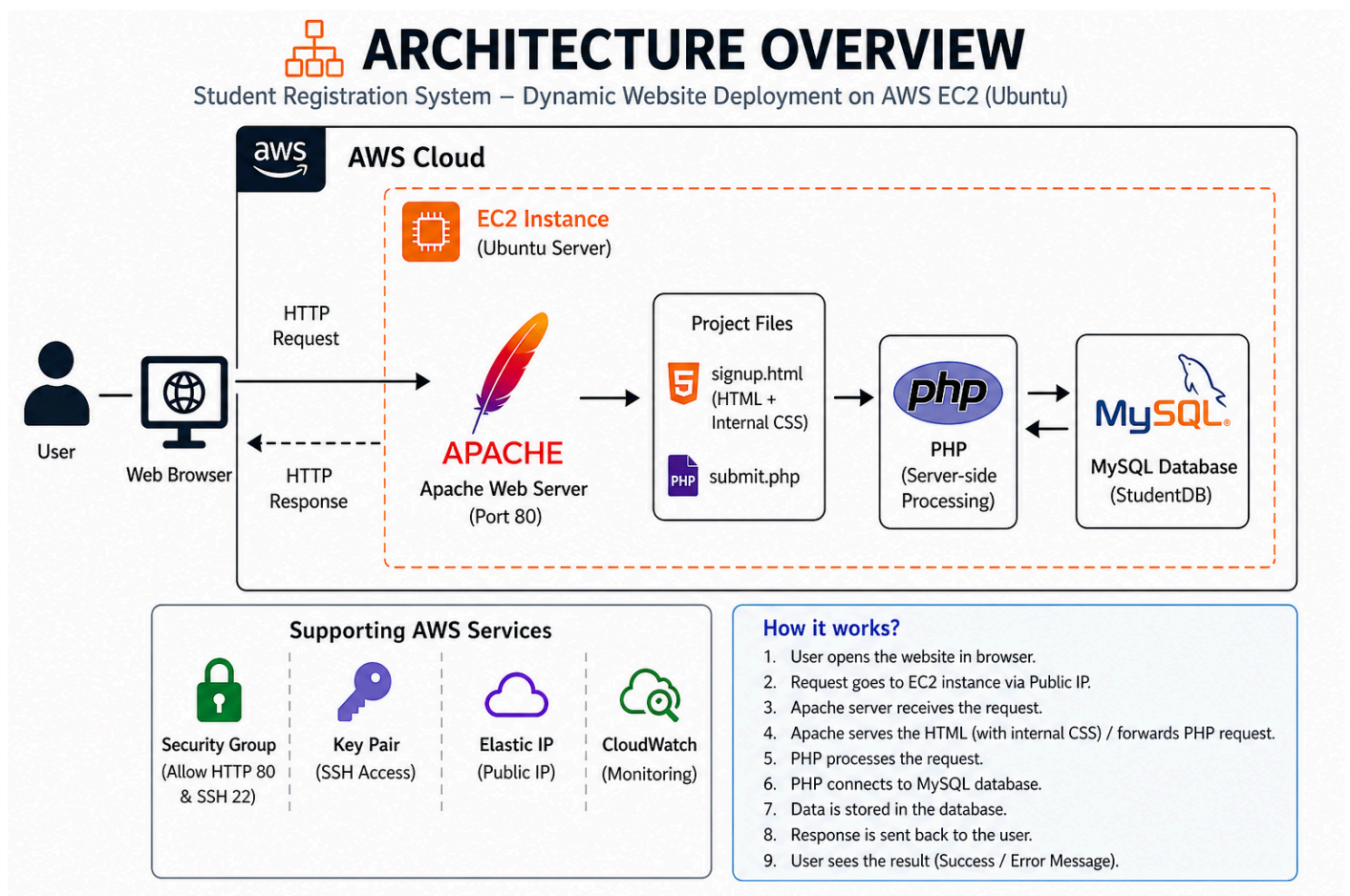
- Signup form using HTML + CSS
- PHP file (submit.php) to handle form submission
- MySQL database to store student data
- Hosted on AWS EC2 (Ubuntu)

Everything connected → frontend → backend → database

Tools & Technologies

- HTML & CSS → form design
- PHP → backend logic
- MySQL → database
- Apache → web server
- AWS EC2 (Ubuntu) → deployment

Architecture Overview



Tech Stack

- Frontend: HTML, CSS, JavaScript
- Web Server: Nginx
- Cloud Platform: AWS EC2 (Amazon Linux)

- Scripting: Bash

⚙️ Deployment Steps

1. Launch EC2 Instance

- AMI: Ubuntu
- Instance Type: t2.micro (Free Tier)
- Security Group: launch-wizard-1
 - SSH (22) → My IP
 - HTTP (80) → Anywhere



The screenshot shows the AWS Management Console interface for the EC2 service. The left sidebar contains navigation links for EC2, including Dashboard, AWS Global View, Events, Instances (selected), Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, and Capacity Manager. The main content area displays the 'Instances (1)' page with a table listing one instance named 'Zomato1' with ID 'i-053a76e5b9a7c5ab9', which is in a 'Stopped' state. The table columns include Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, and Public IPv4. In the top right corner of the console, there is a 'Launch instances' button, which is highlighted by a blue arrow.

aws

Search

[Alt+S]

United States (N. Virginia)

aarti pawar (988530830706)

aarti pawar

EC2 > Instances > Launch an instance

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

Student-Registration-Form

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

aws

Mac

ubuntu

Microsoft

Red Hat

SUSE

debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Summary

Number of instances

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...[read more](#)

ami-0ec10929233384c7f

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

aws

Search

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SUSE

debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type

ami-0ec10929233384c7f (64-bit (x86)) / ami-0ab515046786de9dc (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble image

Architecture

AMI ID

Publish Date

Username

64-bit (x86)

ami-0ec10929233384c7f

2026-03-14

ubuntu

Verified provider

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Cancel

Launch instance

Preview code

file:///C:/Users/Arti/AppData/Local/Temp/omv-pdf-Cif7PU/REDME.print.html

4/13

aws

Search

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United States (N. Virginia)

aarti pawar (988530830706)

aarti pawar

EC2 > Instances > Launch an instance

Instance type

t3.micro
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0104 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour
Free tier eligible
All generations
Compare instance types

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required
my-north-virginia-key
Q | Create new key pair
Proceed without a key pair (Not recommended) Default value
my-north-virginia-key
Type: rsa
vpc-063f507ddb2b1a685
Subnet Info
No preference (Default subnet in any availability zone)
Auto-assign public IP Info

▼ Summary

Number of instances Info
1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-0ec10929233384c7f

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Launch instance
Preview code

aws

Search

[Alt+S]

United States (N. Virginia)

aarti pawar (988530830706)

aarti pawar

EC2 > Instances > Launch an instance

vpc-063f507ddb2b1a685

Subnet Info
No preference (Default subnet in any availability zone)
Auto-assign public IP Info
Enable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group Select existing security group
Common security groups Info
Select security groups
Q | Compare security group rules
launch-wizard-1
VPC: vpc-063f507ddb2b1a685 sg-01663e754e6ad97ec
launch-wizard-2
VPC: vpc-063f507ddb2b1a685 sg-0620b7874a8438a7b
default
VPC: vpc-063f507ddb2b1a685 sg-0620b7874a8438a7b
1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted
Add new volume

▼ Summary

Number of instances Info
1

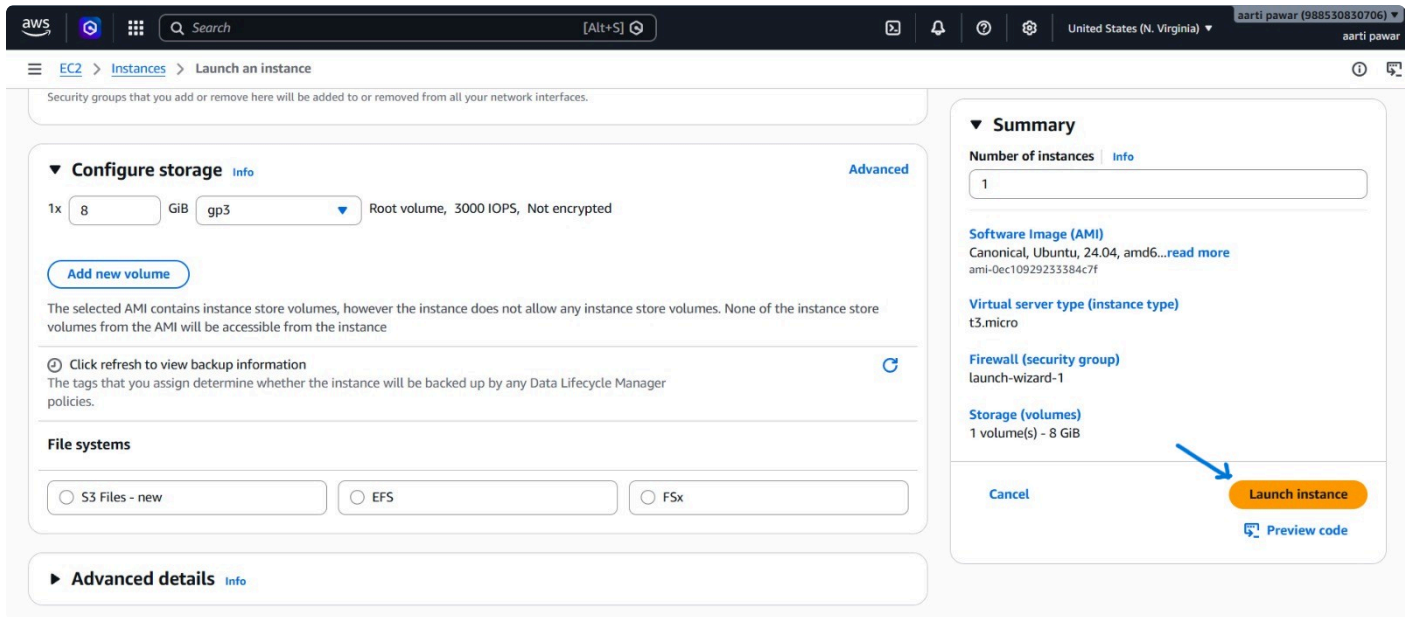
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t3.micro

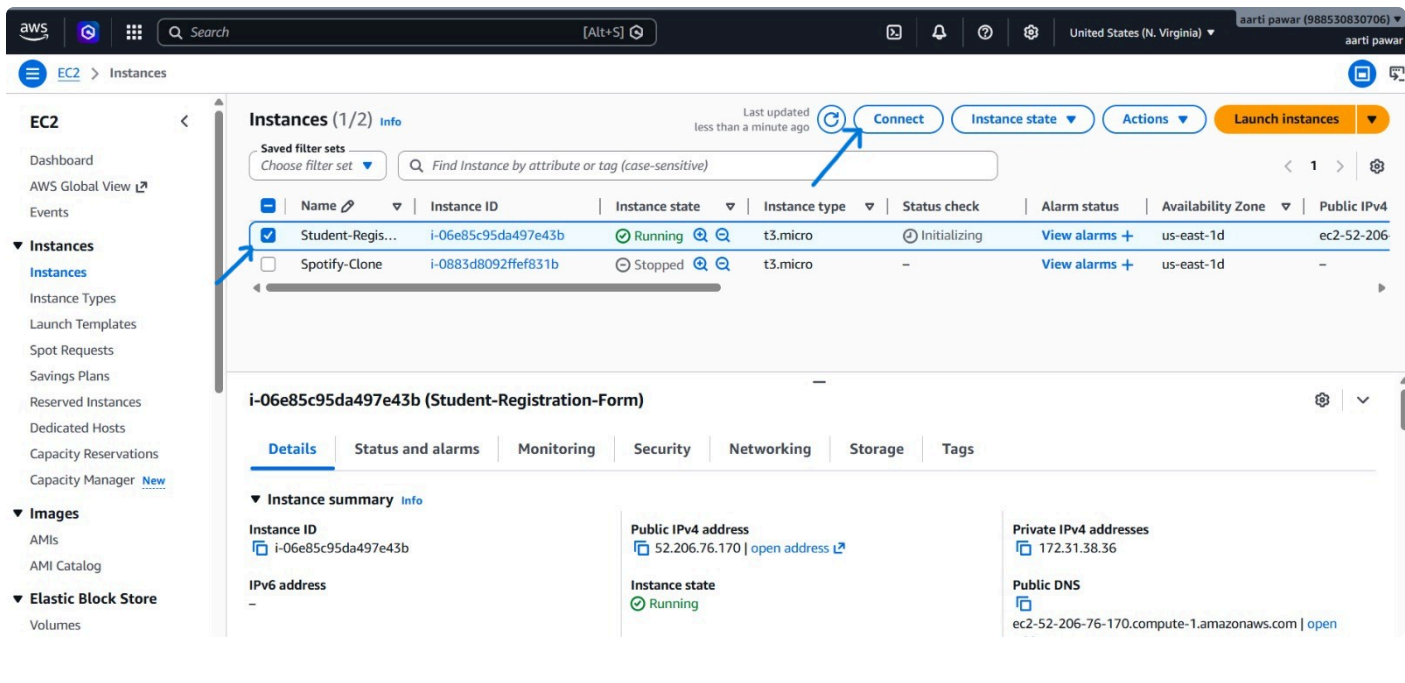
Firewall (security group)
launch-wizard-1

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Launch instance
Preview code



2. EC2 Login using SSH



aws   Search [Alt+S]     United States (N. Virginia) aarti pawar (988530830706) aarti pawar

EC2 > Instances > i-06e85c95da497e43b > Connect to instance

Instance ID  i-06e85c95da497e43b (Student-Registration-Form)	VPC ID  vpc-063f507ddb2b1a685	Security groups  sg-01663e754e6ad97ec (launch-wizard-1)	IAM role -
---	--	--	----------------------

EC2 Instance Connect SSM Session Manager **SSH client** EC2 serial console

Instance ID
 i-06e85c95da497e43b (Student-Registration-Form)

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is my-north-virginia-key.pem
3. Run this command, if necessary, to ensure your key is not publicly viewable.
 `chmod 400 "my-north-virginia-key.pem"`
4. Connect to your instance using its Public DNS:
 `ec2-52-206-76-170.compute-1.amazonaws.com`

Example:
 `ssh -i "my-north-virginia-key.pem" ubuntu@ec2-52-206-76-170.compute-1.amazonaws.com`

 Copy example command to clipboard

username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel

```

ubuntu@ip-172-31-38: ~
Arti@LAPTOP-AUE8RI7V MINGW64 /c/aarti_workspace/ssh.key
$ ssh -i "my-north-virginia-key.pem" ubuntu@ec2-52-206-76-170.compute-1.amazonaws.com
The authenticity of host 'ec2-52-206-76-170.compute-1.amazonaws.com (64:ff9b::34ce:4caa)' can't be
established.
ED25519 key fingerprint is: SHA256:aakU1V1uuQNIXfZmgxQfCFF7aCY/w/2Np5ud719RJ3g
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-52-206-76-170.compute-1.amazonaws.com' (ED25519) to the list of kno
wn hosts.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Sat Apr 25 07:33:22 UTC 2026

System load:  0.01           Temperature:   -273.1 C
Usage of /:   26.3% of 6.71GB Processes:     116
Memory usage: 25%           Users logged in: 0
Swap usage:   0%            IPv4 address for ens5: 172.31.38.36

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
  
```

```
System load: 0.01      Temperature: -273.1 C
Usage of /: 26.3% of 6.71GB Processes: 116
Memory usage: 25%      Users logged in: 0
Swap usage: 0%         IPv4 address for ens5: 172.31.38.36
```

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: `sudo pro status`

The list of available updates is more than a week old.
To check for new updates run: `sudo apt update`

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in `/usr/share/doc/*/copyright`.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "`sudo <command>`".
See "`man sudo_root`" for details.

ubuntu@ip-172-31-38-36:~\$ |

3. Change Hostname

```
sudo hostnamectl hostname Student-Registration
```

LAMP Stack Installation and Configuration

```
sudo apt update
sudo apt install apache2 mysql-server php php-fpm -y
sudo systemctl start apache2 mysql php8.3-fpm
sudo systemctl enable apache2 mysql php8.3-fpm
```

4. Creating Project Files and Database Setup

After installing all required software, I moved to the web server directory:

```
cd /var/www/html
```

Then I created my project files using vim editor:


```
sudo vim signup.html  
sudo vim submit.php
```

In signup.html, I created the form and added CSS inside the head section.

In submit.php, I wrote backend code to connect with MySQL and insert user data.

After that, I opened MySQL and set password for the database:

```
sudo mysql  
alter user root@localhost identified by 'root';  
exit  
sudo mysql -u root -p  
password=root
```

Then I configured database connection details like: username, password, host (localhost) inside submit.php.

Next, I created a database and table to store user data.

```
create database FCT;  
use FCT  
CREATE TABLE users (  
    id INT PRIMARY KEY AUTO_INCREMENT,  
    name VARCHAR(20),  
    email VARCHAR(100),  
    website VARCHAR(255),  
    gender VARCHAR(6),  
    comment VARCHAR(100)  
);  
desc users;
```

```
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affiliates. Other names may be trademarks of their respective
owners.
```

```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
mysql> use FCT
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A
```

```
Database changed
mysql> desc users;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
name	varchar(20)	YES		NULL	
email	varchar(100)	YES		NULL	
website	varchar(255)	YES		NULL	
gender	varchar(6)	YES		NULL	
comment	varchar(100)	YES		NULL	

6 rows in set (0.01 sec)

Issue I Faced

While working on this project, I got confused during MySQL connector installation.

When I was trying on **Amazon Linux**, I used this command:

```
sudo yum install php8.5-mysqlnd.x86_64
```

Later, when I was doing the same setup on **Ubuntu with Apache**, I tried using the same command, but it did not work.

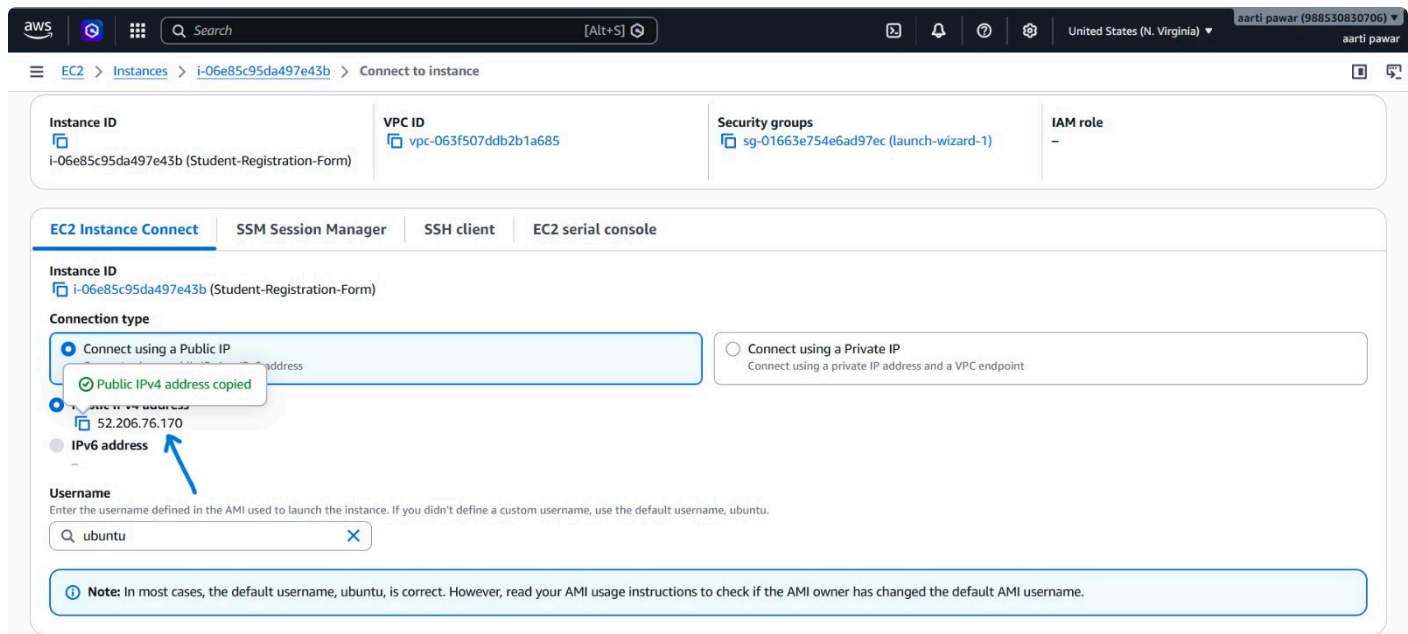
Then I understood that the command is different for Ubuntu, and I used:

```
sudo apt install php-mysql
```

At that time, I got confused between both commands, but after using the correct one, the database connection worked properly.

5. Access The Application

1. Go to AWS EC2 Dashboard
2. Select your running instance
3. Copy the Public IP address

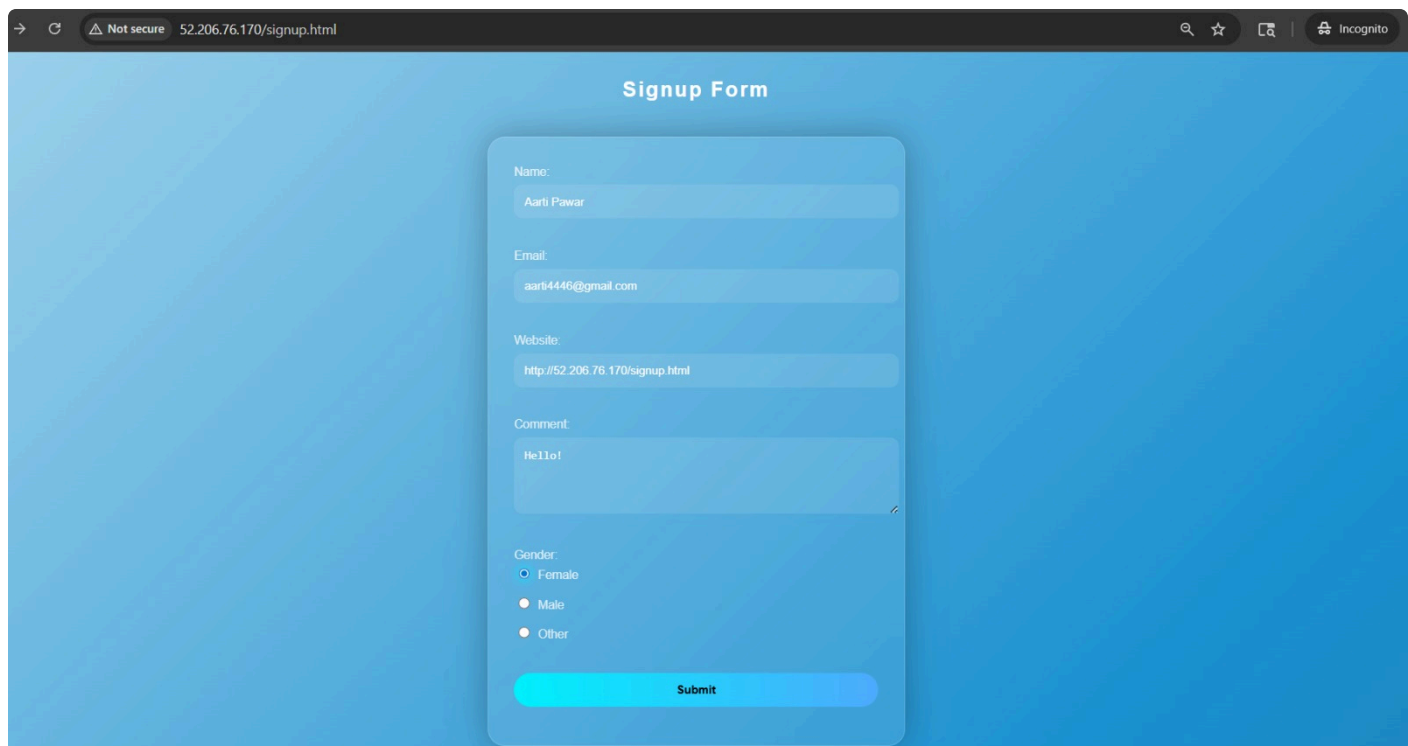


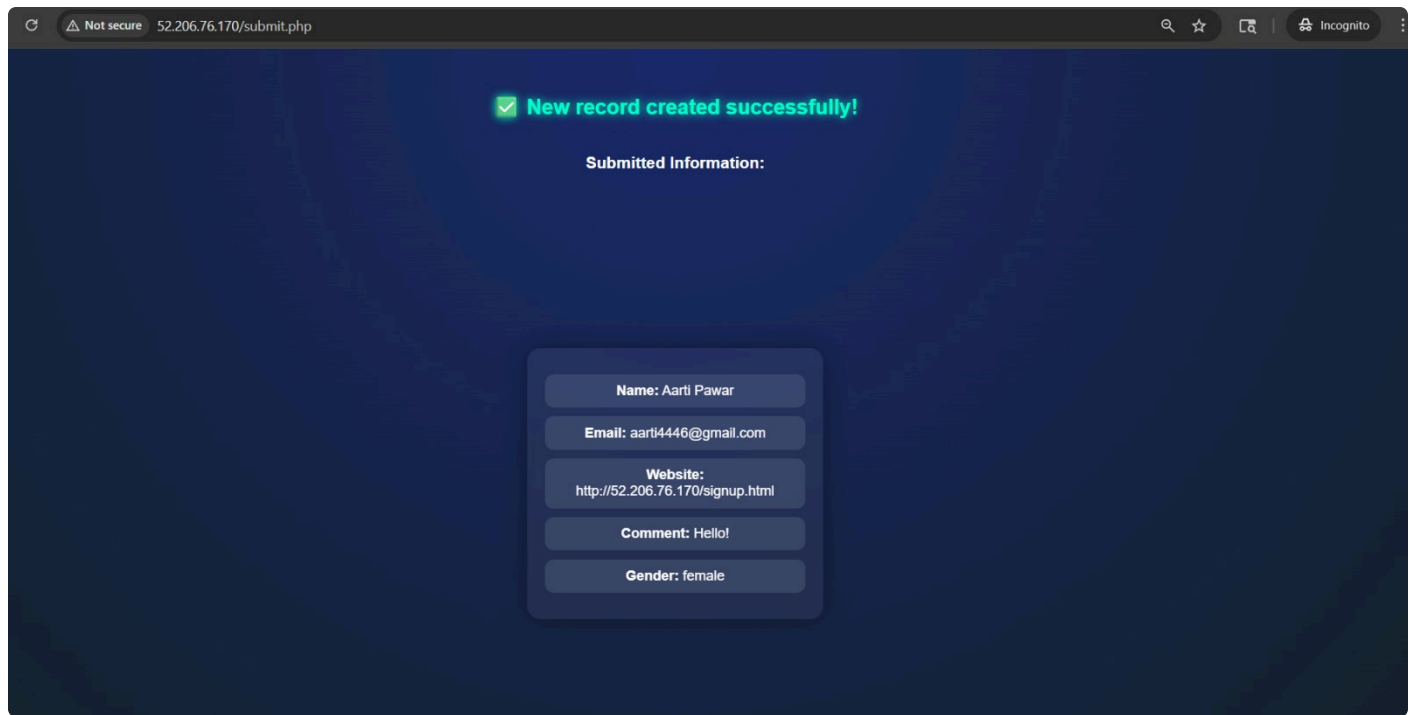
4. Open a new Incognito window in your browser

5. Paste the IP address in the URL like this:

<http:///signup.html>

6. Press Enter to view the website





Key Takeaways

- Learned how to create and connect an EC2 instance using SSH
- Got familiar with basic Linux commands while working on the server
- Understood how Nginx serves static files like HTML and CSS
- Practiced creating and editing files using vim editor
- Used a Bash script to download multiple images in one go
- Understood how a website is accessed using public IP

What Changed After This Project

Before:

- Only static websites

After:

- Understand how data is stored in database
 - Know how PHP connects with MySQL
 - Can deploy dynamic project on cloud
-



What I Want to Do Next

- Add login system
 - Improve UI design
 - Add validation
 - Store password securely
 - Try domain + HTTPS
-



Final Thoughts

This is my first dynamic project where I worked with backend and database.

It is simple, but it helped me understand how real websites handle user data.



Good step forward for me in web development